Five-dimensional attachment and incidence constraints for right-angled hyperbolic polytopes

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Abstract

Every finite-volume entirely ideal right-angled hyperbolic four-polytope occurs as an actual facet of a finite-faced infinite-volume right-angled chamber in hyperbolic five-space. All its original cusps can simultaneously have bounded rectangular four-box links. An exact positive-carrier example has 457 genuine facets and 258 such cusps. This removes a possible one-layer obstruction, without solving finite-volume completion. Separately, the internally proved carrier and second-volume equality theorems imply that an entirely ideal right-angled five-polytope needs at least 56 facets. Published Nikulin–Khovanskii inequalities then restrict the fraction of entirely ideal five-faces in dimensions nine to twelve; in dimensions eleven and twelve it is strictly less than one tenth. These are necessary constraints, not new dimension exclusions.

1 A universal attachment with genuine facets

Use the Lorentz metric of signature $(4,1)$ on the old vector space. Let $Q \subset \mathbb{H}^4$ be a finite-volume entirely ideal right-angled polytope. Choose a strict interior vector $p = (1,0,0,0,0)$ and normalize its ideal vertices as $v = (1,x)$. Write m_i for its outward unit facet normals, so $m_{i0} > 0$. Extend the vector space by an orthogonal unit spacelike vector n.

Theorem 1. *For sufficiently large $L > 1$, the halfspaces with outward normals*

$$n, \quad m_i, \quad h_v = Lv - n$$

define a full-dimensional right-angled chamber in $\mathbb{H}^5$ with exactly $1 + f_3(Q) + f_0(Q)$ genuine facets. Its intersection with $n^\perp$ is exactly Q, and every original ideal vertex has a bounded rectangular four-box cusp link. The chamber has infinite volume.

Proof. Put

$$a = \min_{\langle v,m_i\rangle<0}(-\langle v,m_i\rangle) > 0, \qquad b = \min_{v\neq w}(-\langle v,w\rangle) > 0.$$

Choose $L > 1$ with $La > 1$ and $L^2 b > 2$. Then

$$h_v^2 = 1, \quad \langle n,h_v\rangle = -1, \quad \langle h_v,m_i\rangle \in \{0\}\cup(-\infty,-1), \quad \langle h_v,h_w\rangle = 1 + L^2\langle v,w\rangle < -1 \quad (v \neq w).$$

All wall pairs are orthogonal, asymptotic or ultraparallel, and $p - n/2$ is a strict future timelike interior vector.

On $n^\perp$, the old inequalities define Q and every new inequality is strict at each finite point of Q: future timelike and future null vectors have negative Lorentz product. Thus Q is the actual n facet, with relative-interior witness p.

For an old facet choose a finite relative-interior vector q in Q. For sufficiently small $\epsilon > 0$, $q - \epsilon n$ is timelike, lies on that old wall, and has strictly negative product with all other walls. For a new wall use

$$x_v = v + \frac{p - Ln}{L^2}.$$

It is future timelike with $x_v^2 = -L^{-2} - L^{-4}$, and

$$\begin{aligned} &\langle x_v, h_v \rangle = 0, && \langle x_v, n \rangle = -L^{-1}, \\ &\langle x_v, m_i \rangle = \langle v, m_i \rangle - m_{i0}/L^2 < 0, && \langle x_v, h_w \rangle = L\langle v, w \rangle < 0 \quad (w \neq v). \end{aligned}$$

Every listed wall is consequently genuine.

At each old ideal vertex v, precisely eight walls meet: n, h_v and the six old incident walls. The old walls form three orthogonal opposite pairs whose sums are positive multiples of v; the new pair has sum $n + h_v = Lv$. The eight normals have rank five. The horospherical intersection is therefore a bounded rectangular four-box, while all remaining inequalities are strict near this cusp.

Finally, $u = p - n$ is future null and

$$\langle u, n \rangle = -1, \qquad \langle u, m_i \rangle = -m_{i0} < 0, \qquad \langle u, h_v \rangle = 1 - L < 0.$$

An open ideal spherical patch around its direction and an inward collar satisfy all inequalities. The Klein volume density diverges along such a collar, so the chamber has infinite volume. □

The new opposite normals have the necessary form, rather than being a choice of unrelated local diagrams. The six old cusp normals span a rank-four subspace with null radical $\mathbb{R}v$; its orthogonal complement in the old Lorentz space is $\mathbb{R}v$. A unit normal opposite n and orthogonal to all six must therefore be $\lambda_v v - n$. Its inequality is strict on the interior of Q only when $\lambda_v > 0$. The theorem chooses these parameters simultaneously.

For the frozen sixteen-cell positive-carrier example,

$$f_0(Q) = 258, \quad f_3(Q) = 198, \quad \kappa(Q) = 6, \quad a = 1/25, \quad b = 1/1150.$$

The rational choice $L = 48$ works. The exact certificate checks all 104,196 wall pairs, all 457 strict facet-interior witnesses, all 258 complete cusp links, the strict center and the open ideal escape. No final cone census is needed for these claims. The coordinates are rational in standard Lorentz space; no finite-covolume arithmetic-group claim is made. Additional walls may cut the escape, so finite-volume completion remains open.

2 A necessary 56-facet bound in the ideal five-dimensional case

For an ideal right-angled four-polytope define

$$\Delta(Q) = f_0(Q) - f_3(Q), \qquad \kappa(Q) = 3f_0(Q) - 4f_3(Q) + 24.$$

The separate internal carrier theorem proves the facet identity $\sum_Q \kappa(Q) = 48$ for an ideal right-angled five-polytope. The separate internal second-volume and equality classification implies that $\kappa(Q) > 0$ forces $\Delta(Q) \geq 5$. These geometric theorems are dependencies, not conclusions of the rational calculation below.

Theorem 2. *Every finite-volume entirely ideal right-angled hyperbolic five-polytope has at least 56 facets.*

Proof. Choose a positive-κ four-facet Q. Kolpakov's published facet minimum and equality theorem [1] gives $f_3(Q) \geq 24$, with equality only for the regular 24-cell. That cell has $\kappa = 0$, so $f_3(Q) \geq 25$ and $f_0(Q) = f_3(Q) + \Delta(Q) \geq 30$.

There are $f_3(Q)$ distinct ridge-neighbor facets of the ambient five-polytope. At each of the $f_0(Q)$ cusps of Q, the rectangular four-box link supplies a unique ambient facet opposite Q. Its hyperplane is asymptotic to the hyperplane of Q at precisely that ideal point. It cannot recur at a different cusp, nor equal an orthogonal ridge neighbor. Counting Q itself gives

$$f_4(P) \geq 1 + f_3(Q) + f_0(Q) \geq 56.$$

□

3 Published inputs and mixed-dimensional consequences

A finite-volume right-angled polytope is simple at edges; all its faces are finite-volume and right-angled. We use the published Nikulin–Khovanskii strict average inequality

$$U(n,k,l) = \binom{n-l}{n-k} \frac{\binom{\lfloor n/2 \rfloor}{l} + \binom{\lceil n/2 \rceil}{l}}{\binom{\lfloor n/2 \rfloor}{k} + \binom{\lceil n/2 \rceil}{k}}, \qquad \frac{1}{f_k} \sum_{E \in \Omega_k} f_l(E) < U(n,k,l),$$

for $0 \leq l < k \leq \lceil n/2 \rceil$. We use only $k = 5$, $l = 1, 4$, $n = 9, 10, 11, 12$. Alexandrov [2] and Egorov [3] state the formula and hypotheses explicitly. We also use Dufour's published noncompact bound $f_4(P^5) \geq 1 + \nu_4 = 16$, and the published absence of compact right-angled five-polytopes. The latter removes any compact exception. The original Dufour full article was not accessible for independent proof review; its statements are reported explicitly in the cited primary preprints. No equality classification from that article is assumed.

Let $F = f_5(P)$ and let I count the entirely ideal five-faces. Every five-face has at least 16 facets, and the ideal ones have at least 56. Thus

$$16F + 40I \leq \sum_{E \in \Omega_5(P)} f_4(E) < U(n,5,4)F.$$

Writing $c = 1 - (U(n,5,4) - 16)/40$, we obtain $F - I > cF$. Every nonideal five-face contains a finite vertex, and each finite vertex belongs to $\binom{n}{5}$ five-faces. Hence

$$\binom{n}{5} v_f \geq F - I > cF.$$

An edge is contained in $\binom{n-1}{4}$ five-faces and $2f_1 = nv_f + 2^{n-1}v_i$. Here 2^{n-1} is the number of vertices of the ideal rectangular-box link, not its number of facets. Applying the edge average inequality gives

$$F > av_f + bv_i, \qquad a = \frac{n\binom{n-1}{4}}{2U(n,5,1)}, \quad b = \frac{2^{n-2}\binom{n-1}{4}}{U(n,5,1)}.$$

Since $\binom{n}{5} - ca > 0$, elimination proves

$$v_f > \frac{cb}{\binom{n}{5} - ca} v_i.$$

The exact values are

n	$U(n,5,4)$	$I/F <$	$F >$	$v_f/v_i >$
9	30	7/20	$v_f/2 + 128v_i/9$	3328/45243
10	30	7/20	$v_f + 256v_i/5$	3328/25135
11	20	1/10	$7v_f/2 + 3584v_i/11$	3072/4807
12	20	1/10	$6v_f + 1024v_i$	512/437

The ratio notation presumes $v_i > 0$; the displayed linear inequality also applies when $v_i = 0$. In dimension twelve the necessary relation is $437v_f > 512v_i$. A boundary with at least one tenth of its five-faces entirely ideal is forbidden in dimensions eleven and twelve. These conclusions neither exclude the dimensions nor assert optimality among all published vertex bounds.

4 Proof and computation status

The universal attachment has a full independent internal mathematical review of its proof, producer and standalone checker. Exact generation, fresh replay and the standalone checker passed. The density proof has two independent full internal mathematical reviews, with precise scopes recorded in its companion report. Its script checks rational consequences of explicit internal and published geometric premises. Neither script is an end-to-end formal proof of the geometry. See `docs/right_angled_five_dimensional_constraints_reproduce.md` for immutable inputs, commands and the internal/external lemma map.

References

[1] A. Kolpakov, *On the optimality of the ideal right-angled 24-cell*, Algebraic & Geometric Topology 12 (2012), 1941–1960. https://doi.org/10.2140/agt.2012.12.1941.

[2] S. Alexandrov, *On ideal vertices of right-angled hyperbolic polyhedra*, 2023 preprint. https://arxiv.org/abs/2303.09533.

[3] A. Egorov, *On lower bounds for the number of ideal and finite vertices of right-angled hyperbolic polyhedra in dimensions from 5 to 12*, 2026 preprint. https://arxiv.org/abs/2603.29925.